

BreathMentor Wearable Breathing Monitor

Design Document

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Customer Definition and Challenge:

The BreathMentor is specially designed to tackle the problems related to patients that suffer from GBS (Guillain Barré Syndrome). GBS is a rare auto-immune-mediated neurological disorder that impacts the peripheral nervous system. Roughly 3000-6000 cases of GBS are diagnosed in the continental United States annually. The body's immune system will mistakenly attack the nervous system resulting in nerve damage, weakness and muscle paralysis, loss of reflexes, and difficulty breathing [1].

The client's geographic location is the continental United States. The demographic are those in late stage GBS where they are losing muscle control on their breathing, this is primarily 50-75 years of age. The economic status of the client is open to all, however, the target is towards higher income clients since this electronic device is more advanced than the current breathing [20].

Here are some of the challenges that GBS patients face on a day to day basis. 20% of GBS patients end up developing respiratory failure. Through early training of their breathing, this can be lowered and development slows down [2]. This is why in early stages, it's common for GBS patients to start losing feeling in their body parts that slowly grow over their body as the disorder develops. As it develops, they face paralysis that spreads throughout the body overtime. The physical challenges they face can be physical discomfort, full-time hospitalization, trouble doing everyday tasks such as walking and picking up items, immobilization in advanced cases. While most patients recover, they might still have to do long-term physical therapy to stay alive, this is what this device is trying to help. The device is for patients to train at the comfort of their homes on their own at any time.

The challenge of this product is designing a device that integrates the nucleo board for a hardware and software based application. The aspect of GBS that will be addressed is their need to exercise their intercostal muscles to retain muscle control over their lungs. We will implement this through a trainer that will strap around the patient to measure stretch through their inhalation and exhalation. This can perform a diagnostic and be used for training modules that have the patient follow along a pre-programmed guide with audio output and LCD display output telling them how much to breathe in and out. Their own results will be displayed as one of the values in the LCD to give the user a rough idea of how much effort they have to give and to match with another pair of values in the LCD that displays their progress.

Competitive Landscape:

➤ Respiratory Physiotherapy

- Respiratory physiotherapy are services that hospitals may provide to patients. They involve specialized therapists who work with the patient to improve their lung function by performing personalized exercise routines to improve their respiratory muscle strength. This is a popular treatment that GBS patients undergo [33].
- There are a few shortcomings to this approach. The costs in the United States for private clinics can be unaffordable for many. Without insurance, the physiotherapies range in the price range of \$100-\$500 at private clinics [21]. It can also be inconvenient with patients as they must make the hours to fit the schedule of the hospital and be in person or pay higher fees for the doctors to come to their location. Additionally, availability of these services are geographically limited, patients might not have the option to physiotherapy for their location.

➤ Portable Spirometers

- These are breathing monitors that help patients monitor their lung function and track their breathing. Modern spirometers have digital screens to display results. They are very commonly used by GBS patients to track the progression of the disease and aid in rehabilitation. The information tracked can be used by doctors to provide further diagnoses and prescriptions [34].
- A shortcoming of these devices is that they don't address a way to help improve the user's breathing. Frequent breathing exercises are recommended for GBS patients to slow down the progression of the disease and maintain respiratory muscular control. They are entirely used for tracking the patients inhalation and exhalation and don't provide rehabilitation exercises.

➤ Telemedicine Rehabilitation and Home-Based Breathing Apps

- Telemedicine rehabilitation is a new approach to rehabilitation that gained popularity during the COVID-19 pandemic. It connects the patient and doctor virtually through a device on a video call software where the doctor will instruct the patient in real time for their rehabilitation. There are also home-based breathing apps such as Breathwrk. They provide an at home, in-demand convenience of going through a breathing workout to help patients [35].
- The shortcomings of these approaches are that they don't provide real-time and precise feedback. With telemedicine, since the doctor is observing the patient through a video call, they can't receive accurate information of the patient unlike specialized devices or with in person. With the apps, there isn't much feedback for the patient to know if they are performing the exercises correctly or giving enough effort.

Project Requirements:

➤ Functional Requirements

- The device and specifically the sensor should be capable of sampling data at 7Hz (7 times per second) for accurate data reading and feedback for the patient while minimizing measurement error [3], and be able to store 200 measurements, in line with similar devices like blood glucose monitors [4].
- The device should be able to work with many different body types for inclusivity and to make it compatible with the largest market. The chest circumference range for the first and last percentiles of the male chest circumferences are between 87 cm and 127 cm [5]. The device should be able to reach these lengths while functioning.
- The output devices (audio sensor and LCD display) should have a latency of <100ms. This means that the collection of data, processing of data, and output should be accomplished in <100ms. This will require optimizations in the collection of the outputs, efficiency of the algorithms and processing of data, and in the speed of the output. Visual and auditory output of 5-60ms is detectable by humans and is the standard latency of games. Latency of over 100ms is deemed to be too distracting, which is why the functional requirement for latency should be <100ms [6].

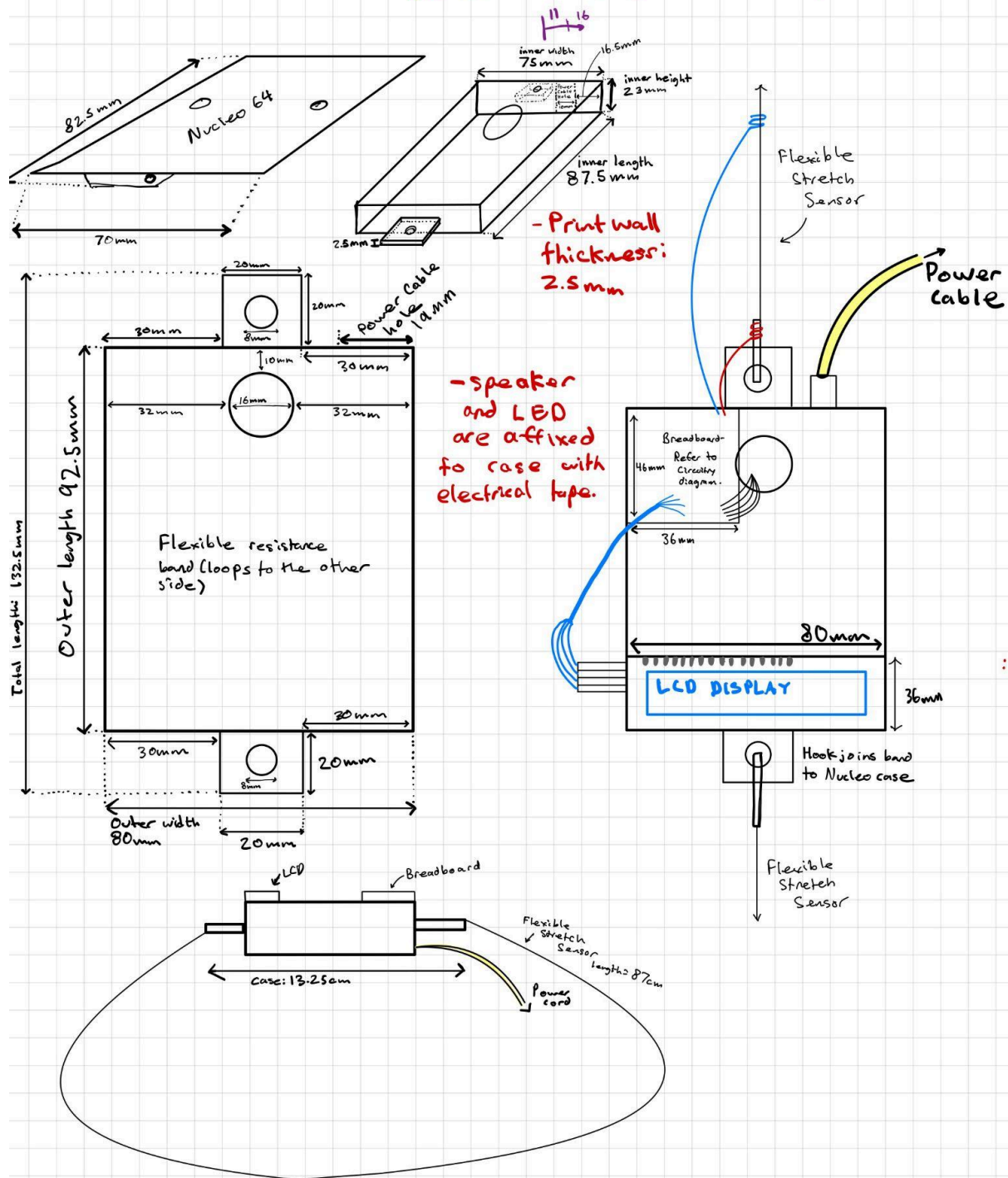
➤ Technical Requirements

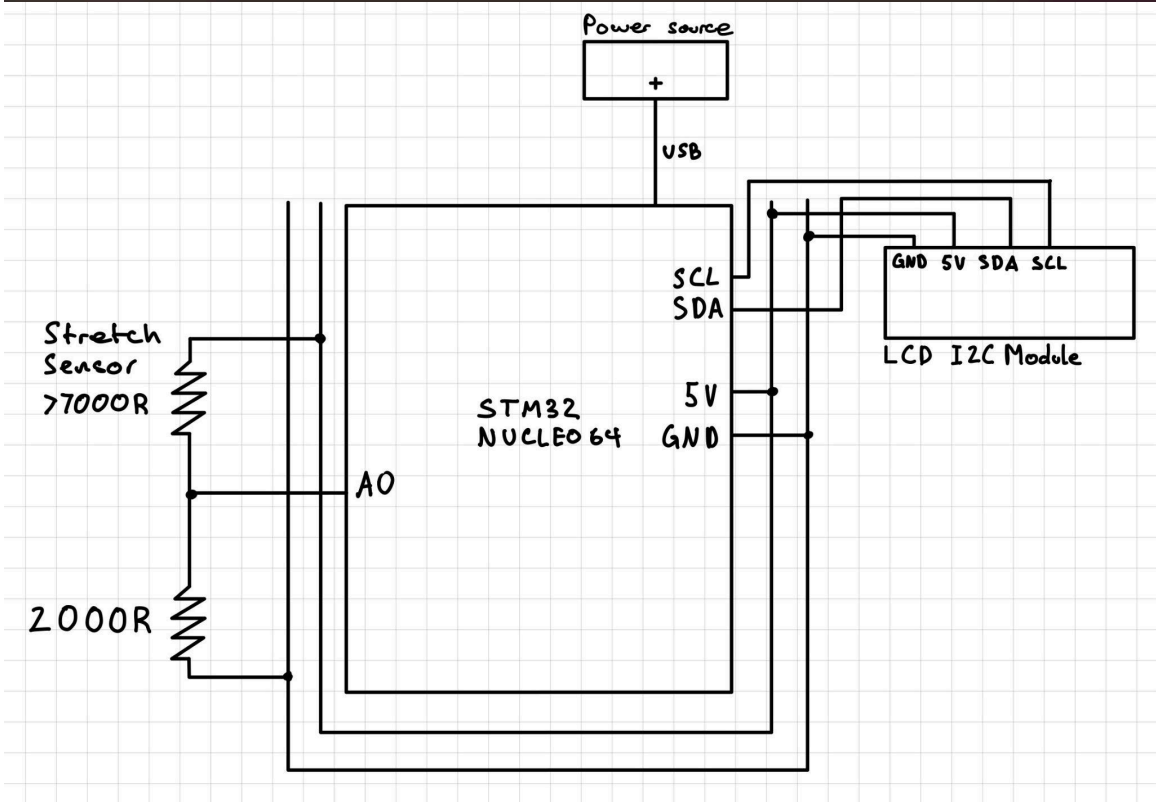
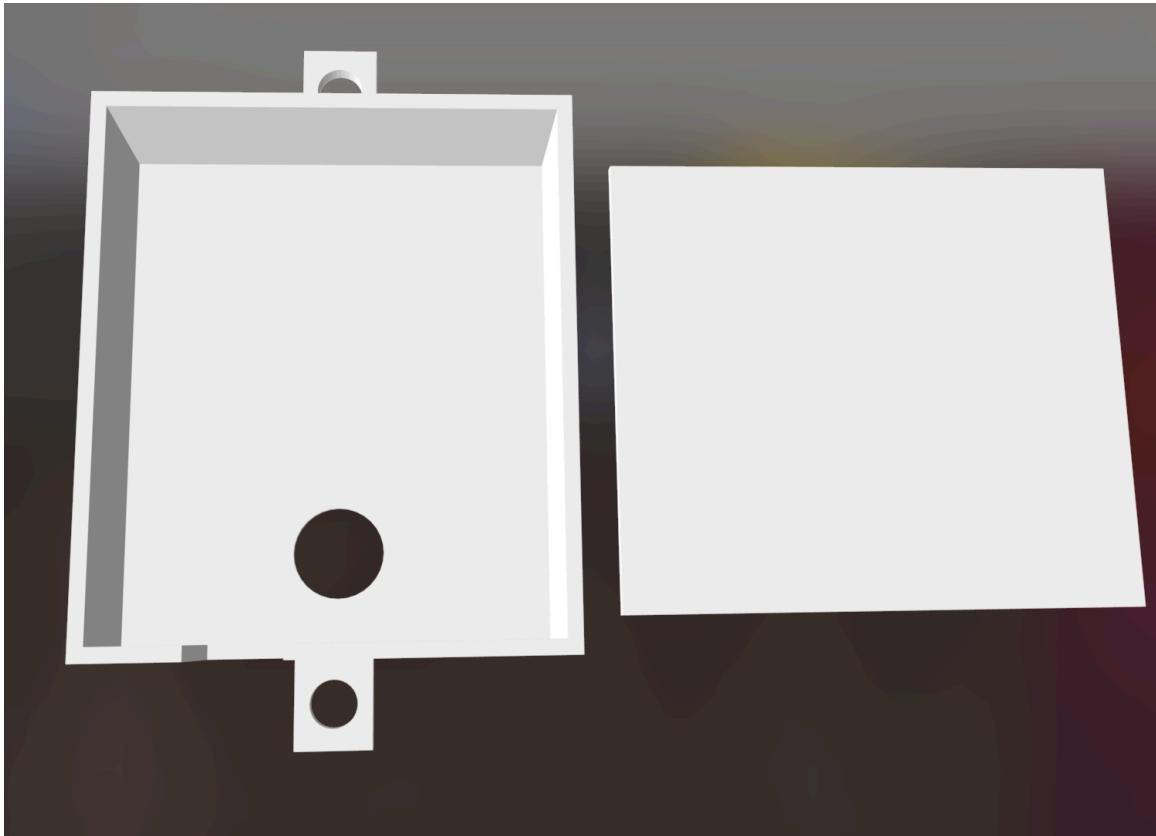
- The system should provide real time feedback, with the stretch sensor being able to provide accurate stretch distance data readings. The sensor should be able to provide data that is accurate up to $\pm 5\%$ (margin of error of 5%) for the amount stretched in centimeters. This not only pertains towards the accuracy of the data readings, but also to minimize fluctuations in the readings when under the same stretch, i.e, the output readings should not be changing by more than $\pm 5\%$ when under the same strain on standby. The sensor output is in resistivity, therefore the units would be in ohms. This is inline with the percent error of medical grade respiratory sensors which traditionally have an absolute margin of error of 3% [8].
- The output of the audio sensor to give instructions should meet the minimum decibel of at least 65 dB. This requirement comes from the fact that the average human conversation is roughly 60 dB [9], therefore, 65 dB should be needed to give instructions for the breathing exercises. This falls inline with the max output of a microcontroller audio speaker which can output 83 dB [10]. The frequency of the speaker should fall in the range of 20Hz to 20kHz as that is the range of frequency for human hearing [11].

Design

- The design is such that the microcontroller is connected to a breadboard with LCD display along with a microcontroller. A stretch sensor is connected end to end via an elastic waistband. The endpoints of the sensor are connected to the microcontroller
- When the sensor stretches, it gives an input of resistivity. This is detectable by the microcontroller. The microcontroller is able to convert the resistivity to force using Young's Modulus, and using Hooke's Law, it is able to convert it to change in distance stretched.
- The change will be showcased with the LCD display. Two sets of two values will be displayed. One set will be set up as an aid/guide for how much the user should be inhaling or exhaling. The other set will be used to represent and reflect the actions of the user, acting as feedback for how much they are actually inhaling or exhaling. The two values will be the distance stretched on the stretch sensor in centimeters, and the other will be the amount of moles of air in moles.
- The speaker will be used to give the user specified instructions for the guided exercise.
- The Nucleo 64 and battery is enclosed within a 3d printed plastic case
- The Nucleo 64 case is connected on one side to a flexible stretch sensor which uses piezoresistivity to measure stretch distance (as the stretch sensor increases in resistivity as length increases).
- The sensor is connected on the other side to a flexible resistance band that loops around to connect to the other side of the Nucleo 64 case. By finding the spring constants of the materials, a ratio of stretch between the band and the sensor will allow for a calibrated exact stretch distance conversion of the device from the increase in resistivity.
- Two wires join from the arduino to one of each side of the stretch sensor. A current will be run through this circuit to measure the resistance, which will be compared against resistance when the resistance band is at rest.

3D Printed Case (inner dimensions)





Mathematical and Scientific Principles:

➤ Hooke's Law

- Hooke's law is a fundamental law in physics that describes the relationship between force applied to a spring or spring-like material and the resulting change in length (deformation) [15].
- $F=kx$
 - For solving for amount stretched: $F/k = x$
 - Where F is the force applied
 - k is the spring constant specific to that spring
 - x is the change in length experienced
- Sample calculation (With arbitrary arguments) ,:
 - We can solve for x by rearranging
 - $x = F/k$
 - We can use force as 36.86N
 - We can use k as 640 N/m
 - Calculation:
 - $x = F/k$
 - $x = 36.86/640$
 - $x = 0.0575 \text{ m}$
- Hooke's law will be used for the stretch sensor which can be approximated as a spring. It will output a specific resistance experienced due to the resistance from being stretched. We can figure out the force that is applied to the sensor and Hooke's law can be used to figure out the length that is stretched by the sensor which will be used for processing.
- The way that Hooke's law will be implemented is through software. The sensors will give a reading as an output in resistance, that resistance could be translated to force using Young's Modulus. Through the simple formula of Hooke's Law and through experimentation to determine the k constant of the sensor, we can determine the distance stretched. This is imperative for readability for the user's end, as outputs of resistance do not serve much context, however changes in the stretch of the sensor presented in units of length provide more context and the product becomes more user-friendly. It will be displayed on the LCD display as one of the two values in each set. One will show how much the user has stretched the sensor and other will show how much the user should be stretching.

➤ Piezoresistivity Law

- Special materials will change their resistivity based on tension applied. Piezoresistivity law relates the mechanical strain, resistivity, and force applied on the material [17].
- $\Delta R = S * (R_0 * \epsilon)$
 - Where ΔR is the change in resistance
 - S is the sensitivity factor in ohms per unit strain
 - R_0 is the initial resistance
 - ϵ is the applied strain
- Sample calculation:
 - We can use this to calculate the sensitivity factor by rearranging the question
 - $S = \Delta R / (R_0 * \epsilon)$
 - The initial resistivity is 350 ohms/inch, we'll be using 5 inches
 - $R_0 = 350(5) = 1750$ ohms
 - The change in resistivity measured is 500 ohms
 - ϵ can be set to 0.124 as an example
 - Calculation:
 - $S = \Delta R / (R_0 * \epsilon)$
 - $S = 500 / (1750 * 0.124)$
 - $S \approx 2.3$ ohms/ ϵ
- The Piezoresistivity law is a way to establish the relationship between the stretch sensor's output and the strain applied on the sensor. This can allow us to figure out the strain experienced by the sensor. The law will be calculated once when creating the project, it will be used to find the sensitivity factor. This can be used with Young's Modulus to convert strain to force. We can substitute strain using the assumption that the strain applied is axial strain (stretch).
- Young's Modulus [18]:
- $F = (\Delta R / R_0) * (1 / S) * (A / L) * E$
 - F : force (in Newtons)
 - ΔR : Change in resistance (in ohms, Ω)
 - R_0 : Initial resistance (in ohms, Ω)
 - S : Piezoresistive sensitivity factor (in ohms per unit strain, Ω/ϵ)
 - A : Cross-sectional area of the material (in square meters, m^2)
 - L : Original length of the material (in meters, m)
 - E : Modulus of elasticity of the material (Pascals, Pa)
- This will all be used in our calculations along with Hooke's Law to determine the stretch of the sensor.

- This is to be implemented with the hardware sensor outputs and software processing. This is the first part to calculate the distance the stretch sensor was stretched to determine how much the user has inhaled and exhaled. Resistance and initial resistance will be calculated throughout the exercise and at the calibration respectively. The other constants are determined through measurements and experimentation with the stretched sensor. The force will be translated to stretch distance using Hooke's Law. Implementation is simple once constants are determined and formula is translated to code.
- Sample calculation:
 - The cord is 0.001m in radius. Therefore the cross sectional area can be found using $A = \pi r^2$
 - $A = \pi(0.001)^2 = 0.000001\pi \approx 0.00000314 \text{ m}^2$
 - The original length of the sensor is 5 inches, which is equivalent to 0.127 m
 - The initial resistivity is 350 ohms/inch, we'll be using 5 inches
 - $R_0 = 350(5) = 1750 \text{ ohms}$
 - The change in resistivity measured is 500 ohms
 - The modulus of elasticity is 12 MPa, which is equivalent to 12,000,000 Pa
 - The sensitivity factor is $2.3 \text{ } \Omega/\epsilon$
 - Calculation:
 - $F = (\Delta R / R_0) * (1 / S) * (A / L) * E$
 - $F = (500/1750) * (1 / 2.3) * (0.00000314 / 0.127) * 12,000,000$
 - $F = 36.86 \text{ N}$

➤ Ideal Gas Law/Pressure Calculations

- The ideal gas law along with pressure calculations can help us determine the amount of air that the patient has inhaled in moles. For this we use a combination of pressure calculations and the ideal gas law.
- Pressure calculations are a means of calculating the pressure from specific parameters [22]. Pressure itself is the measure of force over a certain area. We can calculate pressure with:
 - $P = F/A$
 - P: Pressure (pascals/Pa = N/m²)
 - F is the force applied (Newtons)
 - A is the area (m²)
 - Sample calculation:
 - We can use force as 36.86 N
 - We can approximate the area in contact as being a rectangle. The rectangle has a length of 0.127 m, and a width of the diameter or

the sensor, which would be 2 mm (0.002 m). The area can be found by multiple the two values: $0.002 * 0.127 = 0.000254 \text{ m}^2$

- Calculation:
 - $P = F/A$
 - $P = 36.86 / 0.000254$
 - $P = 145000 \text{ N/m}^2$
 - This value might seem high, but it makes sense since the pressure is applied for m^2 .
- Force can be found with Young's Modulus. The area can be determined through calculations on the stretch sensor predetermined.
- This pressure can then be used for the ideal gas law. The law describes the behavior of ideal gasses through properties of gas [23]:
 - $PV = nRT$
 - In order to find moles: $(PV)/(RT) = n$
 - P: Pressure (Pa)
 - V: Volume (m^3)
 - n: number of moles
 - R: ideal gas constant ($\sim 8.31 \text{ J}/(\text{mol} \cdot \text{K})$)
 - T: absolute temperature of the gas (Kelvin K)
 - Calculations:
 - We can solve for moles by rearranging
 - $n = (PV) / (RT)$
 - We can use P as 14500 N/m^2
 - R is the constant being 8.31
 - The average temperature of the body is 37 celsius, that is equivalent to 310 Kelvin
 - The volume can be approximated as a cylinder. The height of the cylinder is 0.27 m, being the average height of the adult lung. We can use an example circumference of 1.02 m. Finding the volume can be calculated by the cylinder volume formula: $V = \pi r^2 h$. The radius can be found by dividing circumference by 2π : $r = C/2\pi \Rightarrow r = 1.02 / 2\pi = 0.162 \text{ m}$
 - Volume can be found as $V = \pi r^2 h \Rightarrow V = \pi (0.162)^2 (0.27) = 0.022 \text{ m}^3$
 - Calculation:
 - $n = (PV) / (RT)$
 - $n = [14500 * 0.022] / [(8.31 * 310)]$
 - $n = 0.124 \text{ moles}$
- This formula can be used to calculate the approximate change in moles from the patient. The ideal gas constant is already known, the temperature of the gas can be

approximated to the ambient body temperature in kelvin, the pressure is calculated through Young's modulus combined with pressure calculations, and volume can be approximated to be a cylinder. The cylinder would have a circumference of the length of the stretch sensor and band, this will already be calculated through Hooke's law to determine the stretch of the sensor. The height of the cylinder will be approximately 25 cm as that is the average height of a lung [24]. This can be used to determine an approximate change in moles of air the patient inhales. In terms of the implementation to the device, this will be displayed on the LCD display board as one of the two values in each set to the user. One will display the number of moles the user has inhaled and the other will display the number of moles that the user should be inhaling.

Manufacturing Costs

- 3D printed shell component:
 - Requires 3D print filament (material), 3D printer (technology), CAD software (technology). See Below:
- 3D Printer: Fortus 360mc 3D printer
 - Component manufacturer: Stratasys
 - Component Manufacturer's Geographical Location: Headquarters are in Rehovot, Israel [30].
 - Vendor: University of Waterloo Rapid Prototyping Center. Available for use at the E7 Design Bay, University of Waterloo [31].
- 3D Print Filament: ABS-M30 plastic, 0.013"
 - Component manufacturer: Stratasys
 - Component Manufacturer's Geographical Location: Headquarters are in Rehovot, Israel [30].
 - Vendor: University of Waterloo Rapid Prototyping Center. Available for use at the E7 Design Bay, University of Waterloo (4\$/cubic inch) [31].
- Solidworks CAD software
 - Manufacturer: Dassault Systèmes
 - Manufacturer location: Vélizy-Villacoublay, France [32]
 - Sold and distributed by Dassault Systemes
- STM32F401RE Microcontroller:
 - Component Manufacturer: Manufacturer and designers are STMicroelectronics.
 - Component Manufacturer's Geographical Location: Headquarters of STMicroelectronics is Geneva, Switzerland. Location of manufacturing of the

boards is done in factories in Agrate Brianza and Catania (Italy), Crolles, Rousset, and Tours (France), and Singapore [26]

- Vendor: Wstore Waterloo University
- Vendor Geographical Location: South Campus Hall University of Waterloo, Waterloo Ontario
- Cost: The cost of the microcontroller from the vendor was \$34.99 + HST

➤ 1m Flexible Stretch Sensor:

- Component Manufacturer: Adafruit Industries
- Component Manufacturer's Geographical Location: Headquarters and manufacturing plants are located in New York City, USA.
- Vendor: RobotShop
- Vendor Geographical Location: Mirabel, Quebec
- The cost of the flexible stretch sensor was \$14.13 + HST + \$10.80 for Canada Post shipping
 - Out of the 1m of the sensor offered, 5 inches will be used in the final product, so the industrial cost is $\$14.13 * (5/39.37) + \text{HST} + \10.80 , which comes out to \$1.80 + HST + \$10.80

➤ CMS Speaker

- Component Manufacturer: CUI Devices
- Component Manufacturer's Geographical Location: 6405 SW Rosewood St, Tualatin, OR 97062, USA
- Vendor: Digikey.ca
- Vendor Geographical Location: Thief River Falls, Minnesota
- The cost of the component is \$6.20 + HST

➤ TOPERIN Elastic Stretch Band:

- Component Manufacturer: Uline
- Component Manufacturer's Geographical Location: Headquarters located in Pleasant Prairie, Wisconsin, USA. Manufacturing plant in Canada is located in Toronto, Ontario
- Vendor: Uline
- Vendor Geographical Location: Manufacturing plant is in Toronto, Ontario
- The cost is \$19.68 + HST + Free Shipping for 100 yards

- Out of the 100 yards, 0.81 yards (74 cm) will be used, so the industrial cost of the material is $19.68 * (0.81 / 100) = 0.16$. So the industrial cost is \$0.16 + HST + Free Shipping

➤ Copper wires, button, and microcontroller starter kit:

- Provided through the microcontroller kit purchased. The cost of the entire kit will be taken into consideration since most of the components in the kit (wires, breadboard, buttons) will all be used.
- Component Manufacturer: ELEGOO
- Component Manufacturer's Geographical Location: Shenzhen, China
- Vendor: Amazon Canada Mississauga as the marketplace, ELEGOO Official CA as the vendor
- Vendor Geographical Location: Amazon's warehouse location, 6363 Millcreek Drive Mississauga, Ontario L5N 1L8. Vendor's location is Shenzhen, China
- The cost of the kit was \$24.99 + HST + Free Shipping

➤ LCD display

- Component Manufacturer: MikroElektronika
- Component Manufacturer's Geographical Location: Belgrade, Serbia
- Vendor: Digikey.ca
- Vendor Geographical Location: Thief River Falls, Minnesota
- The cost of the component is \$14.36 + HST

➤ Total costs of all components comes to \$104.02, including HST and shipping costs

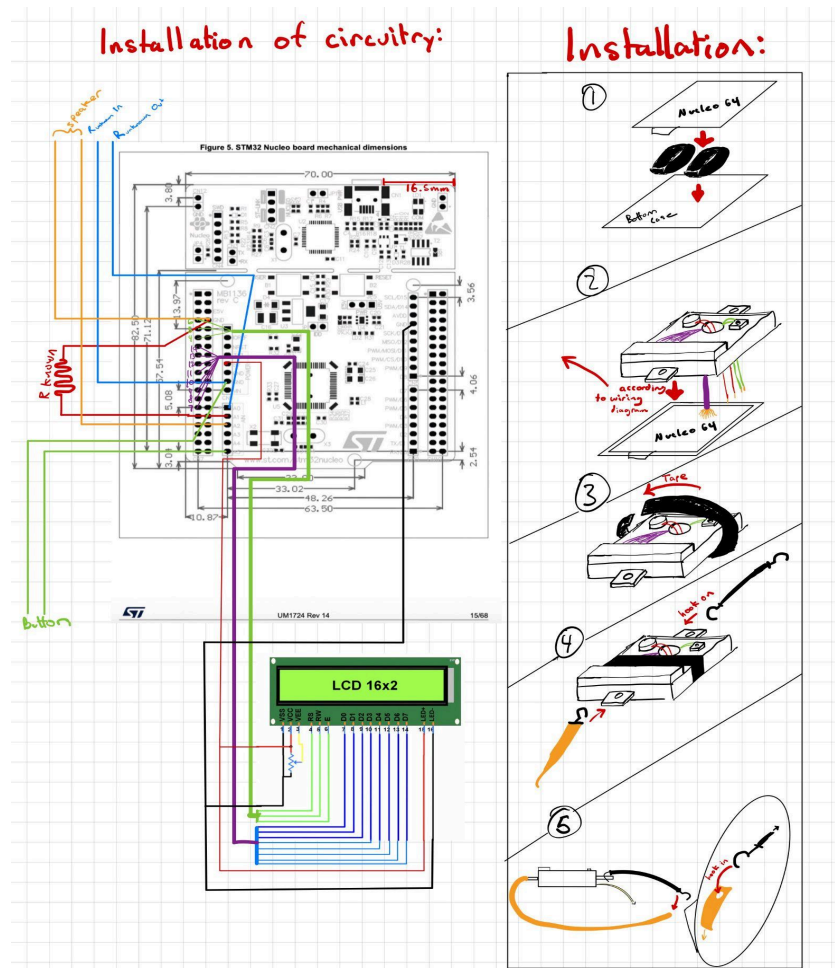
Implementation Costs:

Installation Manual: to construct this device.

1. Parts:

3D printed top case	3D printed bottom case	Power Cord	22 Wires	Stretch Sensor
Electrical tape	Flexible resistance band	LCD arduino part	Speaker Arduino part	Button Arduino Part

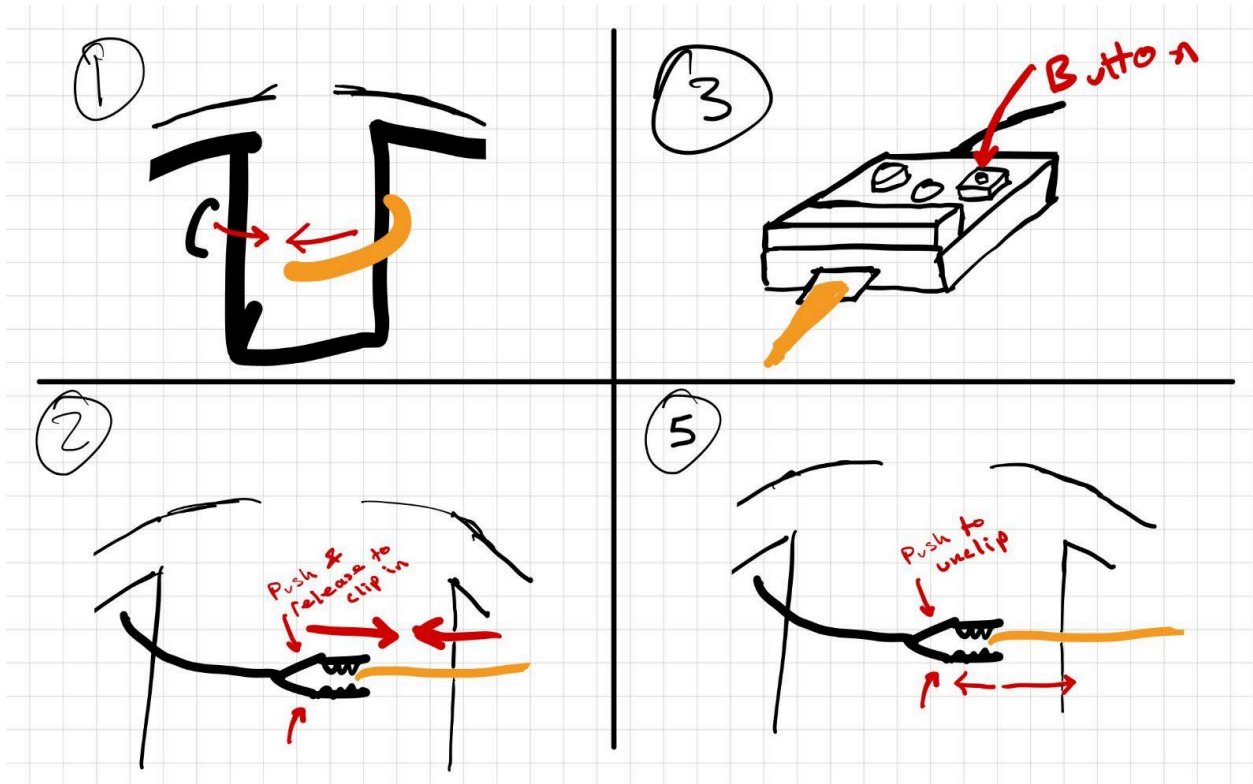
- Tape the 3d printed bottom case to the Nucleo 64
- Connect the wires according to the circuitry diagram below. Especially consider the numbered pins on the LCD display and how they correlate to the numbering of pinholes on the Nucleo 64 wiring diagram.
- Enclose the bottom case and Nucleo 64 assembly with the top case, and tape them together like shown below.
- Hook the stretch sensor to the end of the case with the speaker, and the flexible resistance band to the other side.
- Connect the other end of the resistance band to the other end of the stretch sensor. Installation is now complete.



User Guide:

For everyday use of the device, follow these instructions

1. With the unclicked alligator clips, wrap the stretch sensor your upper chest area, roughly under the armpit
2. Click the other end of the alligator clip to safely secure and close the stretch band around your chest
3. When ready to begin. Click the on button located on the device's microcontroller.
4. Follow the instructions from the speaker. Control your breathing as stated by the instructions. Try to match your breathing with the LCD display, displaying the amount the user should breathe in. The display will also represent the user's current breathing patterns in how much they are inhaling or exhaling so they can aim to match desired values.
5. When completed, unclip one of the alligator clips to open the closed loop, allowing you to put away the device.



Energy Analysis:

- The energy requirements that must be met:
 - The device must not expend more than 30W of power at any point in time and within any component of the design during its operation.
 - This includes all forms of energy, the most relevant to this project are electrical energy (circuitry) and elastic potential energy (stretch sensor and strap)
 - The design must not store or contain more than 500mJ of energy at any point.
 - This includes all forms of energy, the most relevant to this project are electrical energy (circuitry) and elastic potential energy (stretch sensor and strap)
 - Any design component that connects directly to a building electrical supply outlet (110V AC outlet) must be CSA approved.
- Analysis on the design and components
 - Microcontroller
 - From the Section 5 of the STM32 nucleo reference manual [25], these are the following energy specifications of the microcontroller
 - The microcontrollers have an embedded voltage regulator to supply the internal 1.8V digital power domain.
 - The device requires a 2.0V - 3.6V operating and analog supply voltage.
 - During tests, a battery unit won't be used, however, a laptop can be used as a power source. USB2 will be used which has an energy expenditure of 2 watts [27].
 - LCD display output
 - The LCD display that will be used will be the MIKROE-55 HD44780 - LCD Display [28]
 - The max current that is fed through the component is 1.5mA
 - The supply voltage is 3.9V
 - We can calculate wattage through $W = IV$, where W is watts, I is current in amperes, and V is voltage:
 - $W = IV$
 - $W = (0.0015) \cdot (3.9)$
 - $W = 0.00585 \text{ watts}$
 - The component will use 0.00585 watts or 5.85 mW
 - Speaker output
 - The speaker that will be used is the CMS-28588N-L152 speaker from Digikey.ca [29]

- The power rate is 2W, with a maximum total power of 2.5W.
- Stretch Sensor
 - For the stretch sensor, it requires consistent voltage to measure the resistance which is relayed back to the microcontroller. The amount of voltage can be kept at a minimum and still produce a result. The 3.3V pin will be used for the stretch sensor. The amount of watts that the component uses can be calculated with the formula: $W = V^2 / R$, where W is watts, V is volts, and R is the resistance. The resistance of the stretch sensor is 350 ohms per inch when relaxed. 5 inches of the sensor will be used. This means that the maximum wattage the component uses will be:
 - $W = V^2 / R$
 - $W = 3.3^2 / (350 \times 5)$
 - $W = 0.00622$ watts
 - The device will take in roughly 6.22 milliwatts.
 - The amount of mechanical energy of the sensor can be approximated and assumed to be within the limits of the requirements as stated from the authorizing TA. This will be the only energy type of energy that will be stored in the product
- From the components analysis, the total amount of wattage of all components does not exceed the 30W requirement. The amount of energy stored in the device at any one time will be within the limit of the requirement as stated by the supervising TA.

Risk Analysis:

➤ Risk from intended use

- Skin irritation or allergic reaction to the strap
 - There is a possibility that the patient might be allergic or have skin irritation by the material of the stretch sensor and stretch band.
 - The consequences on safety can vary. Generally, this device is meant for older patients and the appeal is that it can be used without any supervision from health professionals. A general skin irritation may cause discomfort or for more severe cases a hospital visit. If the material causes an allergy, the consequences on safety can range from hospital visits or even be fatal if there is no supervision during the time.
- Risk from the strap or the stretch sensor snapping
 - The stretch sensor is not intended to be extended for over double the relaxed length. If the sensor was extended beyond that, then it could

undergo permanent deformation and not work properly or snap. Similarly applies to the stretch band.

- The consequence on safety might injure the patient from the snapping of the sensor or the band. This can be in the form of rashes from the impact, or muscle injuries due to the impact as well. For worst case scenarios, the snapping of the band could hit vital weak areas of the body such as the nose, eyes, or other sensitive areas. For environmental impact, the snapping or the permanent deformation of the sensor would leave the sensor of little use, this can cause extra waste from the sensors.
- Audio output
 - For specific patients, especially the intended demographics of the older patients, some may be more susceptible to shock with sudden loud noises like from the speaker or are sensitive to the sound.
 - The consequence on safety would be discomfort, shock, and headaches. This would be the discomfort and shock they might feel from the speakers turning on suddenly or being too loud for them.

➤ **Risks from incorrect use**

- Discomfort due to strap
 - The strap is not intended for those above and below the top and bottom quartile of chest circumference size. Therefore, if someone who does not fit the size requirement uses the device, they might feel discomfort from the tightness of the device.
 - A consequence on safety due to this is that the patient might feel discomfort on their chest, potentially restricting air flow. For more extreme cases, the pressure might impede their air flow, causing them to lose consciousness or potentially worse. The pressure might also cause an injury, ranging from rashes on the skin to muscle pain from the pressure.
- Using the device laying down or in bed
 - The device is to be used for a period of time during the day where the patient has their full attention in case of mistakes. It is intended to be performed upright. If the device is used laying down, the sensors could malfunction because of the pressure laid on them. The stretch sensor also uses alligator pins, if laid on they could potentially pierce into the body. Laying down also leaves attention from the device, potentially causing it to fall, break, or malfunction.
 - A safety consequence of this is that it could severely injure the patient with cuts and/or bruises. Causing bleeding and potentially hospitalization. An environmental consequence is that the device could break, prompting it to be thrown out and causing e-waste and general waste as well.

➤ **Risks from unintended use**

- Over exertion of the device from intended use
 - The device could be used by those that don't have a risk of GBS, using it for the purpose of exercise instead. Since they are able to perform better, they might overexert the device by performing breathing too quickly or with too much force. The stretch sensor is made out of a material that takes a few seconds to revert back to its relaxed position, risking possible damage or breakage of the sensor and/or the device.
 - A consequence on safety might be potential harm to those that overuse the device for unintended purposes. The breaking of the device could result in electronics parts that can cause harm to the patient. A consequence on safety might be the waste generated from throwing out the device because it is broken.
- Using the strap for unintended applications
 - Using the band unintentionally such as strapping it around the waist, leg, or arm. This is not the intended use and uses the device incorrectly.
 - A safety consequence is that this can injure the user. Using it around a body part that overstretches the device can cause it to snap. Injuries can range from skin irritation to muscle injuries that can cause hospitalizations especially in elderly patients. An environmental consequence is electronic and excess waste due to the device being unusable.

➤ **Possible malfunctions**

- Electronic discharge or exposure of electronics to conductive substances
 - A malfunction could result from the device being used in a very humid or wet environment. The electronics are to be protected from conductive substances that can cause a discharge or conduct the electricity like water.
 - A consequence on safety is that it can electrocute the patient. This can cause discomfort, hospitalization, or even fatal injuries or older patients. A consequence on the environment is that this can become a persistent electrical or fire. It can cause e-waste when the microcontroller and the entire device have to be thrown out.
- Calibration/sensor malfunction
 - A possibility where the sensors send incorrect inputs due to a malfunction or the microcontroller program runs into an unexpected bug causing a malfunction. This would give incorrect outputs and feedback for the user
 - A consequence on safety due to this is that the user might overexert in an effort to match their expected effort from the module. Due to the malfunction, the patient might be getting feedback that they aren't inhaling

or exhaling enough. If the patient overexerts their abilities they can harm themselves from chest pain to going unconscious from exhaling too much because the device incorrectly conveyed they aren't exhaling enough. This would be the worst case scenario for the safety of the patient in this situation.

- LCD output malfunction
 - If there is a malfunction with the connection to the LCD display or with the program, this might cause sudden and irregular flickering. Such malfunction could disorient elderly patients.
 - The consequence on safety due to this could be headache, eye-strain, discomfort, or hospitalization if the flickering disorients the patient such that they fall and injure themselves.

Test Plan:

- Testing the requirement of the device being able to output a minimum of 65 decibels from the output speaker.
 - For the test setup, the microcontroller will be connected to the output audio speaker. The microcontroller will have an audio file that it will output from the speaker. A computer running the website “youlean.co/online-loudness-meter” will be placed 70 cm away from the speaker. The website uses the computer's microphone to measure the decibel readings based off of the sound input. The audio file will run from the speaker.
 - For environmental parameters, the testing room must be at ambient room temperature with little humidity. This is done for safety and to simulate accurate environments where the device can be used commercially. The environment parameters cannot be out of range for the device's optimal operation, such as being too hot to overload the microcontroller. The room must also be entirely silent. This can be measured by the website, with the microcontroller off, the website must detect a maximum of 10 decibels in the ambient room.
 - For test inputs, a pre chosen audio file stored in the microcontroller will be played.
 - For quantifiable measurement, the website gives values in decibels, no conversion needed.
 - For the pass criteria, the website must detect a minimum of 65 decibels for the entire duration of the audio file being played. The audio file will be chosen to be to be a consistent sound.

- Testing the requirement of the device being able to being accurate in its measurement from the inputs
 - For the test setup, the stretch sensor will be connected to the nucleo board and the microcontroller will be connected to a computer that is able to run the code and log in sample outputs. The test will have the stretch sensor being stretched to various controlled lengths specifically chosen from the length requirement. This will be determined by having a measuring tape to verify the actual distance stretched. That information will be fed to the microcontroller and the microcontroller will perform its calculations to convert the resistivity to force then to changes in length. This can be sent to the computer where the values will be outputted to the console.
 - For environmental parameters, the testing room must be at ambient room temperature with little humidity. This is done for safety and to simulate accurate environments where the device can be used commercially. The environment parameters cannot be out of range for the device's optimal operation, such as being too hot to overload the microcontroller.
 - For test inputs, this will just be the stretching of the stretch sensor, the stretch sensor will provide values in resistivity to the microcontroller.
 - For quantifiable measurement, the microcontroller will convert the resistivity in ohms to changes in length in centimeters based on the mathematical and scientific principles programmed. The output value will be in centimeters up to 1 decimal place.
 - For the pass criteria, the output calculated by the microcontroller from the inputs will need to be within $\pm 5\%$ margin of error from the actual value. This can be verified by the measuring tape that will serve as the desired result. Comparing the two values, the output must be less than $\pm 5\%$ in its error.
- Testing the requirement of the device being able to sample data at 7Hz from the sensor for accurate real-time feedback.
 - For the test setup, the stretch sensor will be connected to the nucleo board, allowing the microcontroller to receive inputs. The microcontroller will be connected to a system that can log output data such as a computer or a laptop via ports. This logging device needs to have an already determined frequency and refresh rate much higher than the requirement for the microcontroller, at least >50 Hz. This is done so that the computer won't be the limiting device that gives inaccurate results from the microcontroller. A virtual environment will be set up in the computer that will log the outputs from the microcontroller to the console along with additional software that determines the time delay between each output.

- For environmental parameters, the testing room must be at ambient room temperature with little humidity. This is done for safety and to simulate accurate environments where the device can be used commercially. The environment parameters cannot be out of range for the device's optimal operation, such as being too hot to overload the microcontroller.
 - For test inputs, the stretch sensor will be stretched at various different lengths. This will provide the nucleo board with a variety of input resistances. The software program will convert the resistivity to distance stretched.
 - For the quantifiable measurement standard, the measurement will be done with the latency between each output logged to the console in milliseconds. With each output also being logged with the millisecond difference taken between the current and previous outputs. For it to be under a specified range of frequency in hertz, we can divide 1000 ms by the delay to get the frequency. For this requirement, this means that the millisecond delay must be less than or equal to ~142 ms.
 - The pass criteria of the requirement test is if the observed and measurement value of the refresh and processing of the outputs is within the range of under 7Hz (7 times per second).
- Testing the requirement for the sensor to work for different body types.
- For the test setup, the stretch sensor will be connected to the microcontroller which will be connected to an external computer able to log the outputs given by the microcontroller. The stretch sensor will also be connected end to end on an elastic and stretchable stretch band that is intended to normally be worn around the waist or chest. Two people will be needed, one to stretch the sensor and one to observe the outputs. A measuring tape will be laid out to measure the length of the band.
 - For environment parameters for the room where the test will be conducted, the parameters would be that it is done at standard ambient temperature and pressure, with little humidity. This is done to simulate normal working conditions of the device in commercial applications and for safety reasons for the microcontroller to perform optimally.
 - For test inputs, the stretch band will be stretched at various lengths in the range of the requirement (2 cm to 127 cm). The stretch band will be connected to the microcontroller, allowing it to measure the output.
 - For the quantifiable measurement standard, the measurements will be done in centimeters with the measuring tape. The output given from the device will also be in centimeters calculated through the algorithmic program to convert the resistivity to centimeters.

- For the pass criteria. The stretch band from end to end should be able to stretch in the requirement range of a minimum of 87 cm to a maximum of 127 cm. The output from the microcontroller must also be giving related outputs for the stretch in that range as well. The microcontroller should still be able to detect an accurate input when the band is stretched within the range. The band must not break or show signs of permanent damage during the test and be able to accurately give resistivity inputs within the range with no permanent deformation.
- Testing the requirement for the latency of the output on the LCD display and audio sensor
 - For the test setup, the microcontroller will be connected to the breadboard with the LCD display and audio speaker. This microcontroller will also be connected to a computer that will be running a timestamp software that is able to timestamp and record the times of the inputs and outputs of the microcontroller. The software that will be running is that the microcontroller will be given a command by the computer to represent an example input. The microcontroller must complete its algorithm to convert the input to an output and send a request to the computer and the LCD display and audio speaker. The computer will record the time of the output. The LCD display and audio speaker will be used as controls for any cases where the microcontroller cannot send the inputs properly.
 - When the computer sends an input, the timestamp recording software can record when the input was sent. The microcontroller will receive that input and undergo the program used to calculate the output as it would normally do for real cases. The output will be sent to the computer and the output devices. The computer can use the timestamp recording software to record when the output was received by the computer. The difference between sending the input and receiving an output must be less than 100 ms.
 - For environment parameters for the room where the test will be conducted, the parameters would be that it is done at standard ambient temperature and pressure, with little humidity. This is done to simulate normal working conditions of the device in commercial applications and for safety reasons for the microcontroller to perform optimally.
 - For test inputs, the inputs will be sample inputs given by the computer to the microcontroller. These inputs will be recorded on the computer when they are sent.
 - For the quantifiable measurements, the measurement must be done in milliseconds to match the requirements. Timestamp softwares normally records in microseconds or nanoseconds, this will then need to be converted to milliseconds so that it can be compared.
 - For the pass criteria, software output from the computer about the latency should be <100 ms. Timestamp software can record when the input was read by the

microcontroller and record again for the output signal given to the LCD display and audio speaker. The difference in these records should be less than 100 ms for all cases when testing for the requirement to pass.

References

- [1] W. E. Abara, et al., “Reports of Guillain-Barré Syndrome After COVID-19 Vaccination in the United States,” JAMA Network, Available: [https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2800871#:~:text=Guillain%20Barr%C3%A9%20syndrome%20\(GBS\)%20is%20a%20rare%2C%20immune,annually%20in%20the%20United%20States](https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2800871#:~:text=Guillain%20Barr%C3%A9%20syndrome%20(GBS)%20is%20a%20rare%2C%20immune,annually%20in%20the%20United%20States). [Accessed: September 18, 2023].
- [2] S. E. Leonhard, et al., “Diagnosis and management of Guillain–Barré syndrome in ten steps,” National Library of Medicine, Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6821638/> [Accessed: September 18, 2023].
- [3] F. Benetazzo, et. al., “Respiratory rate detection algorithm based on RGB-D camera: theoretical background and experimental results,” National Library of Medicine, Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4611185/> [Accessed: September 19, 2023].
- [4] “Sinocare Blood Glucose Monitor Safe Accu with Large Memory Capacity, High-cost Performance,” Sinocare, Available: <https://en.sinocare.com/products/sinocare-blood-glucose-monitor-safe-accu> [Accessed September 19, 2023].
- [5] “Average Male Chest Size,” Nutritioneering, Available: <https://www.bodybuildingmealplan.com/average-male-chest-size/> [Accessed: September 19, 2023].
- [6] “Ping, latency and lag: What you need to know,” British Esports, Available: <https://britishesports.org/general-esports-info/ping-latency-and-lag-what-you-need-to-know/#:~:text=Lower%20latency%20is%20ideal%2C%20with,it%20difficult%20to%20play%20properly>. [Accessed: September 19, 2023].
- [7] “Apple Watch Battery,” Apple, Available: <https://www.apple.com/ca/watch/battery/#:~:text=Apple%20Watch%20is%20so%20capable,doing%20a%2060%2Dminute%20workout> [Accessed: September 19, 2023]
- [8] A. Natarajan, et al. “Measurement of respiratory rate using wearable devices and applications to COVID-19 detection,” National Library of Medicine, Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8443549/> [Accessed: September 19, 2023].

- [9] "What Noise Causes Hearing Loss?" Centers for Disease Control and Prevention, Available:
https://www.cdc.gov/nceh/hearing_loss/what_noises_cause_hearing_loss.html#:~:text=A%20whisper%20is%20about%2030,immediate%20harm%20to%20your%20ears.
[Accessed: September 19, 2023].
- [10] A. Raj, "Measure Sound/Noise Level in dB with Microphone and Arduino," Circuit Digest, Available:
<https://circuitdigest.com/microcontroller-projects/arduino-sound-level-measurement>
[Accessed: September 19, 2023].
- [11] D. Purves, G.J. Augustine, D. Fitzpatrick, et al., "The Audible Spectrum," in Neuroscience, 2nd edition. Sunderland, MA: Sinauer Associates, 2001. [Online]. Available:
<https://www.ncbi.nlm.nih.gov/books/NBK10924/#:~:text=Humans%20can%20detect%20sounds%20in,to%2015%E2%80%9317%20kHz.> [Accessed: September 19, 2023].
- [12] "C++ Data Types," W3 Schools, Available:
https://www.w3schools.com/cpp/cpp_data_types.asp [Accessed: September 19, 2023].
- [13] C. David, "Guide to reduce the Arduino Power Consumption," DiyIoT, Available:
<https://diyi0t.com/arduino-reduce-power-consumption/> [Accessed: September 19, 2023].
- [14] "Limit Values," EMF Portal, Available:
<https://www.emf-portal.org/en/cms/page/home/more/electrical-injuries/limit-values#:~:text=A%20touch%20voltage%20of%2050,AC%20or%2060%20V%20DC.>
[Accessed: September 20, 2023]
- [15] The Editors of Encyclopaedia Britannica, "Hooke's Law," Encyclopedia Britannica, Jul. 7, 2023. [Online]. Available: <https://www.britannica.com/science/Hookes-law>.
[Accessed: September 19, 2023].
- [16] The Editors of Encyclopaedia Britannica, "Boyle's Law," Encyclopedia Britannica, Sep. 1, 2023. [Online]. Available: <https://www.britannica.com/science/Boyles-law>.
[Accessed: Sep. 19, 2023].
- [17] "Piezoresistance," ScienceDirect, Available:
<https://www.sciencedirect.com/topics/chemistry/piezoresistance> [Accessed: September 19, 2023].

- [18] “Axial Strain,” ScienceDirect, Available:
[https://www.sciencedirect.com/topics/engineering/axial-strain#:~:text=It%20is%20gene rally%20taken%20at,the%20ultimate%20uniaxial%20compressive%20strength.&text=Average%20Young%27s%20modulus%20is%20determined,2.19\).](https://www.sciencedirect.com/topics/engineering/axial-strain#:~:text=It%20is%20gene rally%20taken%20at,the%20ultimate%20uniaxial%20compressive%20strength.&text=Average%20Young%27s%20modulus%20is%20determined,2.19).) [Accessed: September 19, 2023].
- [19] At what temperature do conventional electronics packages fail?. TWI. (n.d.).
<https://www.twi-global.com/technical-knowledge/faqs/faq-at-what-temperature-do-conventional-electronics-packages-fail#:~:text=Conventional%20electronic%20components%20are%20designed,%C2%B0C%20for%20military%20applications.> [Accessed: September 25, 2023]
- [20] Guillain-Barré syndrome - Louisiana Department of Health. (n.d.-a).
https://ldh.la.gov/assets/oph/Center-PHCH/Center-CH/infectious-epi/Annuals/GullainBarré_LaIDAnnual.pdf [Accessed: October 23, 2023]
- [21] Forbes Magazine. (2023, July 3). *How much does physical therapy cost?*. Forbes.
<https://www.forbes.com/health/body/how-much-does-physical-therapy-cost/> [Accessed: October 23, 2023]
- [22] *Pressure formula*. Third Space Learning. (2023, April 17).
<https://thirdspacelearning.com/gcse-maths/ratio-and-proportion/pressure-formula/#:~:text=Pressure%20is%20calculated%20by%20dividing%20the%20amount%20of%20force%20by%20the%20area.&text=pressure%20%3D%20force%20area%20or%20P,the%20force%20per%20unit%20area.> [Accessed: October 26, 2023]
- [23] Encyclopædia Britannica, inc. (2023, October 19). *Ideal Gas Law*. Encyclopædia Britannica. <https://www.britannica.com/science/ideal-gas-law> [Accessed: October 26, 2023]

- [24] *Lung weight*. Lung Weight - an overview | ScienceDirect Topics. (n.d.). Available: <https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/lung-weight#:~:text=The%20height%20of%20a%20normal,mediastinum%2C%20fill%20the%20thoracic%20cavity>. [Accessed: October 26, 2023]
- [25] *STMicroelectronics*. STMicroelectronics. (n.d.). Available: https://www.st.com/resource/en/reference_manual/rm0091-stm32f0x1stm32f0x2stm32f0x8-advanced-armbased-32bit-mcus-stmicroelectronics.pdf [Accessed: October 30, 2023]
- [26] *STMicroelectronics*. STMicroelectronics. (n.d.). Available: https://www.st.com/content/ccc/fragment/corporate/company_information/company_description/group0/6b/3c/8b/65/8c/91/47/e7/Profile/files/profile.pdf/_jcr_content/translations/en.profile.pdf [Accessed: October 30, 2023]
- [27] *USB charging and Power Delivery: Eaton*. Eaton Website. (n.d.). Available: <https://tripplite.eaton.com/products/usb-charging#:~:text=The%20USB%202.0%20specification%20allows,these%20%22default%22%20power%20levels>. [Accessed: October 30, 2023]
- [28] Mikroe-55 MikroElektronika | Development Boards, kits ... - digikey. (n.d.-d). Available: <https://www.digikey.com/en/products/detail/mikroelektronika/MIKROE-55/4495600> [Accessed: October 30, 2023]
- [29] CMS-28588N-L152 CUI devices | audio products | DigiKey. (n.d.-a). Available: <https://www.digikey.com/en/products/detail/cui-devices/CMS-28588N-L152/6137743> [Accessed: October 30, 2023]
- [30] *Stratasys*. Stratasys. (n.d.). Available: <https://www.stratasys.com/en/> [Accessed: October 31, 2023]
- [31] *Rapid Prototyping Centre*. University of Waterloo. (n.d.). Available: <https://uwaterloo.ca/rapid-prototyping-centre/> [Accessed: October 31, 2023]

- [32] 3ds.com. Dassault Systemes. (n.d.). Available: 3ds.com [Accessed: October 31, 2023]
- [33] *Respiratory physiotherapy*. Physiopedia. (n.d.).
https://www.physio-pedia.com/Respiratory_Physiotherapy [Accessed: October 31, 2023]
- [34] Duong, M., Rangarajan, S., Zaman, M., Nasir, N. M., Seron, P., Yeates, K., Yusufali, A. M., Khatib, R., Tse, L. A., Wang, C., Wielgosz, A., Teo, K., Kumar, R., Avezum, A., Ismail, R., çalık, B. T., Gopakumar, S., Rahman, O., Zatońska, K., ... Yusuf, S. (n.d.). *Differences and agreement between two portable hand-held spirometers across diverse community-based populations in the prospective urban rural epidemiology (PURE) study*. PLOS Global Public Health.
<https://journals.plos.org/globalpublichealth/article?id=10.1371%2Fjournal.pgph.0000141#:~:text=Portable%20spirometers%20are%20commonly%20used,spirometers%20with%20a%20different%20model>. [Accessed: October 31, 2023]
- [35] MD;, G. (n.d.). *Telemedicine in rehabilitation*. Physical medicine and rehabilitation clinics of North America. <https://pubmed.ncbi.nlm.nih.gov/30954160/> [Accessed: October 31, 2023]